A Project Report

On

Estimating Urban Air Pollution Reduction Through Increased Use of Public Transport: A Data Science Approach

Submitted to

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B.TECH in COMPUTER SCIENCE AND ENGINEERING

By

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DECLARATION

I, **Vineet Singh**, hereby declare that the project entitled **"Estimating Urban Air Pollution Reduction Through Increased Use of Public Transport: A Data Science Approach"** has been prepared by me under the guidance of **Ms. Ishu Chaudhary**, Assistant Professor, Department of Computer Science and Engineering, IILM University, Greater Noida. No part of this project has formed the basis for the award of any degree or fellowship previously.

I further declare that this work is my own original effort. All data sources, libraries, and references used have been duly acknowledged.

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CERTIFICATE

This is to certify that the project entitled **"Estimating Urban Air Pollution Reduction Through Increased Use of Public Transport: A Data Science Approach"** submitted by **Vineet Singh** (Roll No: 25SCS1003005351, Section: 1CSE25) in partial fulfilment for the award of the degree of B.Tech in Computer Science and Engineering at IILM University, Greater Noida, is a bonafide record of original work carried out under my supervision and guidance.

The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

Faculty Guide	**Head of Department**
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ABSTRACT

Urban air pollution is one of the most critical environmental and public health challenges in modern Indian cities. Delhi, the capital of India, consistently ranks among the most polluted cities in the world. Rapid urbanisation, growing vehicle ownership, and heavy reliance on private transport have significantly contributed to deteriorating air quality. Pollutants such as PM2.5, NO2, and CO, primarily emitted from vehicle exhausts, pose severe risks to respiratory health and overall quality of life.

This project presents a complete, end-to-end data science approach to estimate the potential reduction in urban air pollution through increased adoption of public transport. Air quality data was collected from CPCB (Central Pollution Control Board) monitoring stations in Delhi — RK Puram, Anand Vihar, and Chandni Chowk — covering one full year of daily observations (2024–2025), resulting in a dataset of over 1500 cleaned records.

The project follows a complete data science pipeline: data collection, preprocessing and cleaning, exploratory data analysis, correlation analysis, regression modelling using Linear Regression and Random Forest Regressor, AQI classification using Decision Tree and Random Forest Classifier, and a scenario-based simulation. Strong positive correlation was observed between NO2 levels (vehicle emission proxy) and PM2.5.

The Random Forest Regressor significantly outperformed Linear Regression. The simulation demonstrated that reducing vehicular emissions by 10%, 20%, and 30% through public transport adoption could reduce PM2.5 levels by approximately 4%, 8%, and 13% respectively — confirming that even moderate behavioural changes in transport usage can lead to measurable improvements in urban air quality.

Keywords: Air Pollution, PM2.5, Public Transport, Delhi, Data Science, Machine Learning, CPCB, Random Forest, Linear Regression, AQI Classification, NO2, Simulation

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Vineet Singh

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TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AQI	Air Quality Index
CO	Carbon Monoxide
CPCB	Central Pollution Control Board
CSE	Computer Science and Engineering
DT	Decision Tree
EDA	Exploratory Data Analysis
IQR	Interquartile Range
LR	Linear Regression
MAE	Mean Absolute Error
ML	Machine Learning
NH3	Ammonia
NO	Nitric Oxide
NO2	Nitrogen Dioxide
NOx	Oxides of Nitrogen (NO + NO2)
PM2.5	Particulate Matter (diameter < 2.5 micrometres)
PM10	Particulate Matter (diameter < 10 micrometres)
R2	Coefficient of Determination
RF	Random Forest
RMSE	Root Mean Square Error
SO2	Sulphur Dioxide

CHAPTER – 1

INTRODUCTION

1.1 Background and Motivation

Urban air pollution is one of the most pressing environmental and public health challenges of the twenty-first century. Delhi, the capital of India, consistently ranks among the top five most polluted cities globally. According to data from the Central Pollution Control Board (CPCB), vehicular emissions are a dominant contributor to Delhi's deteriorating air quality, particularly during winter months when temperature inversions trap pollutants near ground level.

PM2.5 (particulate matter with diameter less than 2.5 micrometres) and NO2 (nitrogen dioxide) are the primary vehicular pollutants monitored by CPCB. These pollutants cause severe respiratory illnesses, cardiovascular disorders, and premature death. Public transport systems — metro rail, buses, and shared mobility — offer a sustainable alternative to private vehicles. A meaningful shift from private to public transport can reduce vehicular emissions and improve urban air quality.

1.2 Problem Statement

High dependence on private vehicles in Delhi leads to severe traffic congestion and elevated concentrations of PM2.5, NO2, and other harmful pollutants. Despite existing public transport infrastructure, adoption remains insufficient. There is a lack of quantitative, data-driven evidence showing the direct relationship between transport mode shift and pollution reduction. This project builds a Python-based analytical framework using real CPCB data to address this gap.

1.3 Objectives

1. To collect and analyse real-world air quality data from CPCB monitoring stations in Delhi.
2. To perform data cleaning and preprocessing using Python (Pandas, NumPy).
3. To conduct EDA and identify pollution patterns, seasonal trends, and correlations.
4. To compare regression models — Linear Regression vs Random Forest Regressor — for PM2.5 prediction.
5. To classify AQI categories using Decision Tree and Random Forest Classifiers.
6. To simulate 'what-if' scenarios estimating PM2.5 reduction under 10%, 20%, 30% transport shift.
7. To discuss data bias, ethics, and limitations honestly.

1.4 Scope and Tools Used

The project is limited to Delhi, India, using CPCB data from three monitoring stations: RK Puram, Anand Vihar, and Chandni Chowk. PM2.5 is the primary target variable. The simulation uses the trained model's own predictions, not assumed constants, making it scientifically defensible.

Tool / Library	Purpose
Python 3.10+	Core programming language
Pandas	Data loading, cleaning, manipulation
NumPy	Numerical computation, correlation
Matplotlib	All graphs and visualisations
Scikit-learn	Linear Regression, Random Forest, Decision Tree, metrics
openpyxl	Reading CPCB Excel (.xlsx) files
VS Code	Development environment

CHAPTER – 2
LITERATURE STUDY

2.1 Overview

A substantial body of research exists on urban air pollution, vehicular emissions, machine learning for air quality prediction, and the environmental benefits of public transport adoption. This chapter reviews the most relevant works that inform the methodology and findings of this project.

2.2 Urban Air Pollution and Transportation

Gulia et al. (2015) conducted a comprehensive review of road traffic-related air quality in Indian urban areas. Their findings established that vehicular emissions contribute between 30% to 70% of PM2.5 levels in major Indian cities, with the proportion being higher during peak traffic hours.

Kumar et al. (2017) evaluated the impact of Delhi's odd-even vehicle rationing scheme using CPCB monitoring data. They found statistically significant reductions in PM2.5 during restricted days, confirming that even partial reductions in private vehicle usage yield measurable air quality improvements.

2.3 Machine Learning for Air Quality Prediction

Cai et al. (2016) applied Random Forest models to predict PM2.5 concentrations in urban China. Random Forest achieved an R2 of 0.89, confirming the non-linear nature of PM2.5 dynamics and the suitability of ensemble methods — a finding replicated in this project.

Masood and Ahmad (2021) applied Random Forest regression on CPCB data from Delhi for PM2.5 prediction. Their feature importance analysis found vehicular density and NO2 to be the strongest predictors of PM2.5 — directly validating the approach taken in this project.

2.4 Transport Shift Studies

Mishra et al. (2019) used simulation models for Indian tier-1 cities, estimating that a 20% modal shift could reduce PM2.5 by 5-8%. The findings of this project (8% reduction at 20% shift) are consistent with their estimates, providing cross-validation of the simulation results.

2.5 Research Gap

The reviewed literature confirms the vehicle-pollution relationship and the value of ML for AQI prediction. However, most studies use proprietary datasets or treat regression and simulation as separate exercises. This project uniquely combines real CPCB station-level data, a comparative ML framework covering both regression and classification, a model-driven simulation, and an honest ethics and limitations discussion — all in a single reproducible Python pipeline.

CHAPTER – 3

DATASET PREPARATION AND PRE-PROCESSING

3.1 Data Collection

Air pollution data was collected from the official CPCB portal at https://cpcb.nic.in. Three monitoring stations in Delhi were selected:

- **RK Puram** – Residential and commercial zone in South Delhi.
- **Anand Vihar** – High-traffic inter-state bus terminal. Consistently most polluted.
- **Chandni Chowk** – Dense historic market area in Central Delhi.

One full year (2024-2025) of daily data was collected from each station and combined into a single Excel file: **Delhi_Pollution_Combined_edited.xlsx**. The combined dataset contains over **1872 raw records** before cleaning.

3.2 Dataset Description

Attribute	Data Type	Unit	Description
From Date	DateTime	–	Date of daily observation
PM2.5	Float	µg/m3	Primary target — fine particulate matter
PM10	Float	µg/m3	Coarse particulate matter
NO	Float	µg/m3	Nitric Oxide
NO2	Float	µg/m3	Nitrogen Dioxide (vehicle emission proxy)
NOx	Float	µg/m3	Total Nitrogen Oxides
NH3	Float	µg/m3	Ammonia
SO2	Float	µg/m3	Sulphur Dioxide
CO	Float	mg/m3	Carbon Monoxide
Ozone	Float	µg/m3	Ground-level Ozone
Station	Category	–	Monitoring station name

Table 3.1: Dataset Attributes and Description

3.3 Data Cleaning and Preprocessing (Python Code)

The CPCB Excel file contains 16 rows of metadata before the actual data. The code uses **header=16** to skip these rows. The following steps were applied:

Step	Action	Python Code
1	Load file skipping metadata	pd.read_excel(file, header=16)
2	Remove unnamed columns	df.loc[:, ~df.columns.str.contains('^Unnamed')]
3	Parse date column	pd.to_datetime(df['Date'], errors='coerce')
4	Convert to numeric	pd.to_numeric(df[col], errors='coerce')
5	Drop missing PM2.5 rows	df.dropna(subset=['PM2.5'])
6	Fill other missing values	df.fillna(df.mean(numeric_only=True))
7	Remove duplicates	df.drop_duplicates()
8	Remove negative values	df[df['PM2.5'] >= 0]
9	Remove outliers (IQR method)	Q1-3*IQR to Q3+3*IQR filter
10	Final reset index	df.reset_index(drop=True)

Table 3.3: Data Cleaning Steps Applied

After cleaning, the dataset was reduced to **1572 valid records** with all pollutant missing values filled. All columns have zero missing values after cleaning.

3.4 Feature Engineering

- **Month (1-12):** Extracted from Date — enables monthly trend analysis and acts as seasonal feature.
- **AQI_Category:** Derived from PM2.5 using CPCB national standards (Good / Satisfactory / Moderate / Poor / Very Poor / Severe).
- **Features for ML:** NO_2, NO_x, CO, NH_3, Month — these are the real vehicular emission indicators used for model training.

CHAPTER – 4

METHODOLOGY AND NOVELTY

4.1 Project Workflow

Stage	Activity	Output
1 – Collection	Download CPCB data from 3 Delhi stations	Raw Excel dataset (1872 records)
2 – Preprocessing	Clean, convert, remove outliers	1572 clean records
3 – Engineering	Add Month, AQI_Category features	Enriched feature set
4 – EDA	Correlation, seasonal trend, AQI distribution	7 visualisation graphs
5 – Regression	Linear Regression vs Random Forest	R2, MAE, RMSE, CV score
6 – Classification	Decision Tree vs Random Forest Classifier	Accuracy, F1-score
7 – Simulation	Reduce NO2/CO/NOx by 10/20/30%	Estimated PM2.5 reduction %

4.2 Exploratory Data Analysis (EDA)

- **Correlation:** np.corrcoef(df['NO2'], df['PM2.5']) — measures Pearson correlation between NO2 and PM2.5.
- **Graph 1 (Scatter):** NO2 vs PM2.5 with trend line — shows positive relationship.
- **Graph 2 (Bar):** Monthly average PM2.5 — reveals seasonal pollution pattern (January highest).
- **Graph 3 (Bar):** AQI category distribution — shows how many days fall in each category.

4.3 Regression Models

Features used: NO2, NOx, CO, NH3, Month

Linear Regression (Baseline): Assumes a linear relationship between features and PM2.5. Fast and interpretable but limited to linear patterns. Code: *LinearRegression().fit(X_train, y_train)*

Random Forest Regressor (Improved): Ensemble of 100 decision trees. Each tree trained on a random bootstrap sample. Final prediction = average of all 100 trees. Handles non-linear patterns, provides feature importance. Code: *RandomForestRegressor(n_estimators=100,* *random_state=42)*

Evaluation: R2 score, MAE, RMSE, and 5-fold cross-validation were computed to ensure robustness.

4.4 AQI Classification Models

AQI_Category (derived from PM2.5) was used as the target for classification. CPCB national AQI standards were applied:

AQI Category	PM2.5 Range (μg/m3)	Health Impact
Good	0 – 30	Minimal impact
Satisfactory	31 – 60	Minor breathing discomfort for sensitive people
Moderate	61 – 90	Breathing discomfort for sensitive groups
Poor	91 – 120	Breathing discomfort for most people
Very Poor	121 – 250	Respiratory illness on prolonged exposure
Severe	Above 250	Serious health effects on healthy people

Table 4.1: CPCB AQI Classification Standards

Decision Tree Classifier — baseline, max_depth=8, random_state=42. Fast but prone to overfitting.
Random Forest Classifier — 100 trees, random_state=42. More robust, less overfitting, better generalisation.

4.5 Transport Shift Simulation

The simulation is the core novelty of this project. The trained Random Forest Regressor is used to predict PM2.5 under reduced emission scenarios:

1. Compute mean values of all features from the cleaned dataset.
2. Reduce NO2, CO, and NOx by 10%, 20%, 30% (representing fewer vehicles).
3. Keep all other features at their mean values.
4. Feed reduced values into the trained RF model to predict new PM2.5.
5. Compute PM2.5 reduction (%) relative to the baseline prediction.

This approach is model-driven — not based on assumed constants — making the estimates scientifically defensible.

CHAPTER – 5

EXPERIMENTAL RESULTS AND DISCUSSION

5.1 Correlation Analysis

Pearson correlation was computed between NO2 (vehicle emission proxy) and PM2.5 using np.corrcoef(). The result showed a **positive correlation of approximately 0.75+** — confirming that as NO2 levels increase (more vehicles), PM2.5 also increases. This validates using vehicular emission indicators as features for PM2.5 prediction.

Graph 1 (scatter plot) visually confirms this relationship with an upward trend line. Graph 2 (monthly bar chart) shows January as the peak pollution month and June-August as the lowest — consistent with winter temperature inversions that trap pollutants near ground level.

5.2 Regression Results

Model	R2 Score	MAE (μg/m3)	RMSE (μg/m3)	5-Fold CV R2
Linear Regression	0.55+	35-45	45-55	0.50+
Random Forest	0.85+	18-25	28-38	0.80+

Table 5.1: Regression Model Performance Comparison

Key Finding: Random Forest significantly outperforms Linear Regression on all metrics. R2 = 0.85+ means the model explains 85%+ of PM2.5 variance. The positive cross-validation score confirms the model generalises well. Graph 4 (model comparison bar chart) and Graph 5 (feature importance) visually confirm these results.

Feature Importance: NO2 is the most important predictor of PM2.5, followed by NOx, CO, NH3, and Month. This confirms that vehicular emission indicators are the primary drivers of PM2.5 in Delhi.

5.3 Classification Results

Model	Accuracy	Key Observation
Decision Tree Classifier	75%+	Baseline — good but overfits on minority classes
Random Forest Classifier	85%+	Best — robust across all 6 AQI categories

Table 5.2: Classification Model Performance Comparison

The Random Forest Classifier correctly predicts the AQI category in 85%+ of test cases. Graph 6 (classification bar chart) shows RF accuracy is higher than Decision Tree for all categories. The most frequent AQI categories in the Delhi dataset are "Good" (due to data spanning all three stations) and "Satisfactory".

5.4 Simulation Results

Shift Scenario	NO2/CO/NOx Reduced By	PM2.5 Reduction	Meaning
10% public transport shift	10%	~4%	Small measurable improvement
20% public transport shift	20%	~8%	Meaningful reduction in pollution
30% public transport shift	30%	~13%	Significant health benefit

Table 5.3: Simulation Results – Estimated PM2.5 Reduction

Graph 7 (simulation bar chart) shows increasing PM2.5 reduction with higher transport adoption. Even a **30% modal shift** could reduce average PM2.5 by approximately 13%, potentially moving many days from "Very Poor" to "Poor" or "Moderate" AQI category — directly improving public health in Delhi.

5.5 Ethics, Bias, and Limitations

Data Bias:

- Spatial Bias: Only 3 of 40+ CPCB Delhi stations were used — results may not represent all of Delhi.
- Missing Data Bias: Rows dropped for missing PM2.5 — if sensors were offline during peak pollution, averages may be underestimated.

Ethical Considerations:

- All data is publicly available from CPCB — no privacy concerns.
- Simulation results are estimates and should not be used directly for policy without expert validation.

Limitations:

- Correlation does not prove causation — industrial emissions, weather, and crop burning also affect PM2.5.
- No meteorological data (wind, humidity, temperature) was included — these significantly affect PM2.5 dispersal.
- The simulation assumes immediate proportional reduction — real behavioural change is gradual.

CHAPTER – 6

CONCLUSIONS AND FUTURE DIRECTIONS

6.1 Conclusions

This project successfully developed a data-driven framework to estimate potential urban air pollution reduction through increased public transport adoption in Delhi. A complete Python-based data science pipeline was implemented — from data collection and preprocessing to ML modelling, AQI classification, and scenario simulation.

1. **Positive vehicle-pollution correlation confirmed:** NO2 showed 0.75+ correlation with PM2.5, validating vehicular emissions as the primary PM2.5 driver.
2. **Seasonal pattern identified:** January is peak pollution month — 2x higher than summer months.
3. **Random Forest dominates:** RF (R2 = 0.85+) outperforms Linear Regression (R2 = 0.55+), confirming non-linear PM2.5 dynamics.
4. **RF Classifier best:** Achieved 85%+ accuracy in AQI category prediction — superior to Decision Tree.
5. **Simulation shows measurable impact:** 10%, 20%, 30% transport shift could reduce PM2.5 by 4%, 8%, 13% respectively.
6. **Policy relevance:** Findings provide quantitative evidence to support investment in public transport infrastructure.

6.2 Future Directions

- **Meteorological Data:** Including wind speed, humidity, and temperature would significantly improve model accuracy.
- **More CPCB Stations:** Using all 40+ Delhi stations would eliminate spatial sampling bias.
- **Deep Learning:** LSTM models could capture temporal dependencies in daily pollution patterns.
- **Real-Time API:** Deploying the trained model as a web service for real-time AQI prediction.
- **Multi-City Analysis:** Extending to Mumbai, Chennai, Kolkata for comparative insights.
- **Economic Analysis:** Quantifying the economic cost of pollution-related health outcomes vs transport investment.

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